WORKED SOLUTION

# STEP II 2002, Question 8

Pure • First-Order Differential Equations

**STEP II 2002, Question 8**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q8

## The key idea

Each equation is the simplest first-order linear equation there is, $y' = p(x)\,y$, so on any interval where $p$ is given by a single formula the solution is a constant times $\exp(\int p\,\mathrm{d}x)$. The catch is that the problem is split at $x = 0$: we get a separate arbitrary constant on each side, fixed on one side by the given value of $y$ and on the other side only by insisting that the two pieces meet at $x = 0$. The first part is a rehearsal of exactly this, with an absolute value hidden in the second.

## Solution

### The first pair of equations

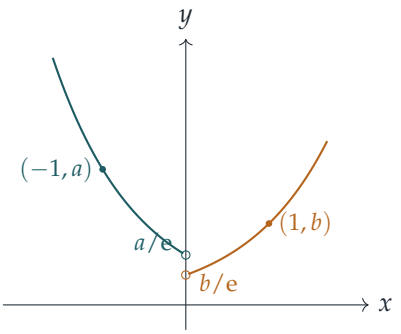
For $x < 0$, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -y$ gives $y = A\mathrm{e}^{-x}$ for some constant $A$. (To see that there are no other solutions, note that $\frac{\mathrm{d}}{\mathrm{d}x}(y\mathrm{e}^{x}) = \mathrm{e}^{x}(y' + y) = 0$, so $y\mathrm{e}^{x}$ is constant on the interval $x < 0$.) The condition $y = a$ at $x = -1$ gives $A\mathrm{e} = a$, so

$$y = a\,\mathrm{e}^{-x-1} \qquad (x < 0).$$

Similarly, for $x > 0$ we have $y = B\mathrm{e}^{x}$, and $y = b$ at $x = 1$ gives $B = b/\mathrm{e}$, so

$$y = b\,\mathrm{e}^{x-1} \qquad (x > 0).$$

As $x \to 0^{-}$, $y \to a/\mathrm{e}$; as $x \to 0^{+}$, $y \to b/\mathrm{e}$. In general these differ, and the curve jumps at the $y$-axis. The sketch shows a typical case with $a, b > 0$ and $a > b$: on the left a decaying exponential, falling as it approaches the axis, and on the right a growing exponential, with open circles marking the two (different) limiting values.




There is no jump exactly when the two limits agree, $a/\mathrm{e} = b/\mathrm{e}$, that is, when

$$a = b.$$

In that case the curve is $y = a\,\mathrm{e}^{|x|-1}$, a symmetric "V"-like curve with a corner at $x = 0$.

**The equation with $|\mathrm{e}^x - 1|$**

Since $\mathrm{e}^x > 1$ exactly when $x > 0$,

$$|\mathrm{e}^x - 1| = \begin{cases} \mathrm{e}^x - 1, & x \geq 0, \\ 1 - \mathrm{e}^x, & x < 0. \end{cases}$$

So the equation splits into two ordinary linear equations, just as before.

**For $x > 0$:** $y' = (\mathrm{e}^x - 1)y$. An antiderivative of $\mathrm{e}^x - 1$ is $\mathrm{e}^x - x$, so $\frac{\mathrm{d}}{\mathrm{d}x}\left(y\,\mathrm{e}^{-(\mathrm{e}^x - x)}\right) = 0$ and

$$y = C\exp(\mathrm{e}^x - x).$$

At $x = 1$, $y = \mathrm{e}^{\mathrm{e}}$, so $C\,\mathrm{e}^{\mathrm{e}-1} = \mathrm{e}^{\mathrm{e}}$, giving $C = \mathrm{e}$. Hence

$$y = \exp(\mathrm{e}^x - x + 1) \qquad (x \geq 0).$$

**For $x < 0$:** $y' = (1 - \mathrm{e}^x)y$. An antiderivative of $1 - \mathrm{e}^x$ is $x - \mathrm{e}^x$, so

$$y = D\exp(x - \mathrm{e}^x).$$

**Ask yourself** There is no data point with $x < 0$. What fixes $D$?

Continuity at $x = 0$ does. From the right, $y \to \exp(1 - 0 + 1) = \mathrm{e}^2$. From the left, $y \to D\exp(0 - 1) = D/\mathrm{e}$. Equating, $D = \mathrm{e}^3$, so

$$y = \exp(x - \mathrm{e}^x + 3) \qquad (x < 0).$$

Putting the two together,

$$y = \begin{cases} \exp(\mathrm{e}^x - x + 1), & x \geq 0, \\ \exp(x - \mathrm{e}^x + 3), & x < 0. \end{cases}$$

Notice that this time there is not even a corner at $x = 0$: both formulas give $y'(0) = 0$, matching $|e^0 - 1|\,y = 0$. Since $y > 0$ and $|e^x - 1| \geq 0$, the solution is increasing everywhere, with a stationary point of inflection at $x = 0$, where $y = e^2$.

#### The limits

**(i)** For $x > 0$,

$$y\exp(-e^x) = \exp(e^x - x + 1 - e^x) = e^{1-x} \to 0 \quad \text{as } x \to +\infty.$$

So the limit is 0: the factor $\exp(-e^x)$ cancels the double-exponential growth exactly, leaving a decaying $e^{1-x}$.

**(ii)** For $x < 0$,

$$y\,e^{-x} = \exp(x - e^x + 3 - x) = \exp(3 - e^x) \to e^3 \quad \text{as } x \to -\infty,$$

since $e^x \to 0$. So the limit is $e^3$. In other words, for large negative $x$ the solution behaves like $e^3e^x$, as you would expect: there $|e^x - 1| \approx 1$ and the equation is nearly $y' = y$.

## Checks

- Differentiate the answer: for $x > 0$, $y' = (e^x - 1)y$, and for $x < 0$, $y' = (1 - e^x)y$. At $x = 1$, $\exp(e - 1 + 1) = e^e$.
- Numerically, stepping the equation from $y(1) = e^e \approx 15.154$ back to $x = 0$ with a Runge–Kutta method gives $y(0) \approx 7.38906 = e^2$, and continuing to $x = -5$ agrees with $\exp(x - e^x + 3)$ to six significant figures.
- For limit (ii), at $x = -10$ we get $ye^{-x} \approx 20.0846$, close to $e^3 \approx 20.0855$; for (i), at $x = 10$, $e^{1-x} \approx 1.2 \times 10^{-4}$.

## Common slips

- Using the condition at $x = 1$ to fix the constant on *both* sides of $x = 0$. The two intervals are separate problems, and the left-hand constant comes only from continuity.
- Dropping the modulus and solving $y' = (e^x - 1)y$ for all $x$; this gives the wrong curve for $x < 0$ and the wrong limit in (ii).
- Writing $\int (e^x - 1)\,dx = e^x - x$ but then $\exp(e^x - x) = e^{e^x}/x$ (it is $e^{e^x}e^{-x}$).

## Beyond STEP

The equation $y' = |e^x - 1|\,y$ has a coefficient that is continuous but not differentiable at $x = 0$. The existence and uniqueness theorem for $y' = f(x, y)$ (Picard–Lindelöf) only needs $f$ continuous in $x$ and Lipschitz in $y$, so here a single condition fixes the solution

on the whole line, and continuity at $0$ is automatic. In the first part the right-hand side itself jumps at $x = 0$, which is why an extra condition was needed to join the pieces. Boyce and DiPrima, *Elementary Differential Equations*, discusses these existence and uniqueness questions for first-order equations.

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